

Question Paper

Exam Date & Time: 24-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: ICT 3123/IT_3123 SOFTWARE ENGINEERING

(SPL: INFORMATION TECHNOLOGY)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) A university wants to automate its online course registration system. The system allows students to log in, view available courses, register for new courses, and receive confirmation messages upon successful registration. The system also checks for prerequisites and seat availability before confirming enrollment. (5)
- Design a detailed Sequence Diagram for the course registration process described above, clearly showing the interaction among objects such as Student, CourseCatalog, RegistrationSystem, and Database. Ensure your diagram reflects message order, conditional checks, and system responses.

Course Outcome: "CO2" / Cognitive Level: "Create"

- 1B) A software development team is designing an online payment gateway that interacts with multiple external modules-such as User Authentication, Order Processing, Transaction Handler, and Notification Service. During design review, it is observed that changes in one module often require modifications in several others, causing high maintenance effort. (3)
- Analyze the design of this system to identify the type of coupling exhibited among its modules. Explain how this level of coupling affects system maintainability and propose one improvement to reduce coupling.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 1C) Differentiate between unit testing, integration testing, and system testing strategies. Explain when each should be applied in the software development lifecycle. (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 2A) An organization is creating an e-commerce application architecture that integrates inventory management, user authentication, payment services, and customer notifications. During system design, the architect must decide how different components will interact and how the system will handle both dynamic content and performance optimization. (5)
- Analyze the system requirements and determine suitable architectural views (logical, process, and deployment) for this application. Explain how these views help in addressing system performance and integration challenges.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 2B) An online food delivery system allows users to: (3)
- Begin in the Order Browsing state.
 - Move to Order Placed after confirming the selection.

- Transition to Preparing Order, then Out for Delivery, and finally Delivered.
 - A user can Cancel before the order is prepared, returning to Order Browsing.
- Construct a UML State Transition Diagram illustrating the order life cycle in the food delivery system. Include states, events, and transitions, ensuring all conditional paths are represented.

Course Outcome: "CO2" / Cognitive Level: "Create"

- 2C) In a student management system, the "StudentModule" performs different tasks such as adding student details, generating reports, and calculating exam grades. (2)
- Analyze whether this module has high or low cohesion and explain your answer briefly.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 3A) Consider the following code segment: (5)

```
function processGrade(score) {
  if (score >= 90) {
    grade = "A";
  } else if (score >= 80) {
    grade = "B";
  } else if (score >= 70) {
    grade = "C";
  } else if (score >= 60) {
    grade = "D";
  } else {
    grade = "F";
  }
  if (grade == "A" || grade == "B") {
    status = "Pass";
    if (score >= 95) {
      honor = "Distinction";
    }
  } else if (grade == "C" || grade == "D") {
    status = "Pass";
  } else {
    status = "Fail";
  }
  return [grade, status, honor];
}
```

Construct the Control Flow Graph (CFG) for this code segment and calculate the cyclomatic complexity using all methods. Design a comprehensive set of test cases that achieve basis path coverage. Justify how your testcases ensure thorough testing of all independent paths.

Course Outcome: "CO4" / Cognitive Level: "Create"

- 3B) A software project faces the following risks: 70% probability of key developer leaving (cost impact: \$12,000), 40% probability of requirement changes (cost impact: \$8,000), and 25% probability of technology integration issues (cost impact: \$20,000). (3)

Analyze these risks by calculating risk exposure for each using the Risk Exposure formula. Develop a comprehensive RMMM (Risk Mitigation, Monitoring, and Management) strategy for the highest priority risk, including specific mitigation steps, monitoring indicators, and contingency plans.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 3C) Distinguish between functional and non-functional requirements in software engineering. Analyze how proper requirements specification impacts software design and testing phases with relevant examples. (2)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 4A) A password validation system has the following requirements: (5)
- Password length: 8-20 characters
 - Must contain at least one uppercase letter
 - Must contain at least one lowercase letter
 - Must contain at least one digit
 - Must contain at least one special character (!@#%&^*)
 - Cannot contain spaces
 - Cannot be identical to the last 5 passwords
- Apply equivalence class partitioning and boundary value analysis techniques to design a comprehensive test suite for this password validation system. Create test cases for both valid and invalid equivalence classes, identify boundary values for the length constraint, and evaluate how your test cases achieve maximum coverage with minimum redundancy.

Course Outcome: "CO4" / Cognitive Level: "Create"

- 4B) An e-learning platform allows students to enroll in courses, submit assignments, take quizzes, and track progress. The system involves actors: Student, Instructor, and Administrator. Develop a comprehensive use case model for this e-learning system that includes at least 5 major use cases with proper relationships. Create an activity diagram for the "Submit Assignment" use case that shows the complete workflow including validation steps and error handling. (3)

Course Outcome: "CO2" / Cognitive Level: "Create"

- 4C) Compare Agile methodology with Unified Process (UP) in terms of their approach to requirements handling, iteration management, and risk management. Analyze the scenarios where each methodology would be most appropriate. (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 5A) A library management system manages books, members, authors, and transactions. The system tracks book borrowing, returns, reservations, and member information. Consider the following relationships: a book can have multiple authors, a member can borrow multiple books, books can be reserved by multiple members, and the system maintains borrowing history. (5)
- Design a comprehensive class diagram for this library management system that includes at least 5 classes with appropriate attributes, methods, and relationships with multiplicities.

Course Outcome: "CO2" / Cognitive Level: "Create"

- 5B) Analyze the role of Software Configuration Management in coordinating team-based software development. Evaluate how version control, baseline management, and change control contribute to project success. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5C) Differentiate between Aspect-Oriented Software Development and Formal process models. (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_3123:SOFTWARE ENGINEERING

#	Course Outcome	Description	Marks
1	1	CO 1	4
2	2	CO 2	18
3	3	CO 3	10
4	4	CO 4	12
5	5	CO 5	6
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_3123:SOFTWARE ENGINEERING

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	0	0(0%)
4	Analysis	9	24(48%)
5	Create	0	0(0%)
6	Evaluate	6	26(52%)
Total		15	50 (100%)